

Ravenstack Customer 360

SMART Framework Questions

Data Analytics

Data Science / ML

CHURN & RETENTION

Q1

What is the monthly churn rate per subscription plan tier during the pilot period, and which tier shows the highest churn concentration?

Every month, some customers cancel their subscription — that's churn. This question asks us to count how many customers left, grouped by the plan they were on (e.g. Free, Starter, Pro). We then want to know: which plan loses the most customers?

Why it matters

If most churn is coming from one specific plan, the business can focus retention efforts there before the official launch — rather than treating all customers the same.

Specific

Broken down by plan tier, not just overall churn

Measurable

Churn rate = $\text{churned} / \text{total active that month} \times 100$

Achievable

All needed data is in the raw CSVs

Relevant

Directly impacts pricing and plan design at launch

Time-bound

Limited to the pilot window only

Data Analytics

📄 subscriptions.csv, churn_events.csv

Q2

Which referral acquisition channels produce customers with the lowest 90-day churn rate during the pilot period?

Customers come from different sources — partner links, social media ads, direct invites. This question asks: which of those sources brings in customers who actually stay for at least 90 days?

Why it matters

Before launch, the business needs to decide where to spend its marketing budget. Channels that bring in customers who leave quickly are not worth the investment.

Specific

Measurable

Achievable

Relevant

Time-bound

Channel-level analysis, 90-day survival window	Churn rate % per channel within 90 days of signup	Referral channel and signup date both available	Guides pre-launch acquisition budget decisions	90 days from each customer's signup date
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Data Analytics

 accounts.csv, churn_events.csv

Q3

Can we predict, with $\geq 75\%$ precision, which active accounts will churn within the next 30 days using subscription history and support ticket behavior?

Instead of reacting after a customer leaves, can we predict who is about to leave — before they do? This question asks us to build a machine learning model that looks at a customer's past activity and gives a risk score: 'this account is likely to cancel in the next 30 days.'

Why it matters

If we know who is at risk, the team can reach out with a discount, a solution, or extra support — before that customer cancels. The 75% precision target means: out of every 100 accounts the model flags, at least 75 should actually churn. That keeps alert fatigue low.

Specific	Measurable	Achievable	Relevant	Time-bound
30-day prediction horizon, binary churn label	Precision $\geq 75\%$, also tracking recall and AUC	Labeled historical churn data exists for training	Core ML deliverable — enables proactive retention	Model predicts churn 30 days into the future

Data Science / ML

 subscriptions.csv, support_tickets.csv, churn_events.csv

REVENUE & GROWTH

Q4

What percentage of accounts upgraded to a higher plan tier within 60 days of signup, and does this vary significantly by industry segment?

Some customers start on a basic plan and then upgrade to a more expensive one as they see the value. This question measures: how often does that happen within the first 60 days? And are certain industries (e.g. tech vs. healthcare) more likely to upgrade than others?

Why it matters

If certain industries upgrade a lot, sales and onboarding can be tailored to push those industries toward premium plans from day one — maximising revenue early.

Specific Upgrade events within 60 days, by industry	Measurable Upgrade rate = upgrades / total signups × 100	Achievable Plan change history captured in subscription data	Relevant Shapes upsell strategy and onboarding flows	Time-bound 60-day window from each account's signup date
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Data Analytics

📄 accounts.csv, subscriptions.csv

SUPPORT & OPERATIONS

Q5

Can we forecast weekly support ticket volume for the next 4 weeks, segmented by plan tier, to anticipate staffing needs at launch?

Based on how many support tickets were created each week during the pilot, can we predict how many will come in during the 4 weeks after launch? And can we break that down by plan tier, since different tiers may have very different volumes?

Why it matters

If launch brings a spike in tickets, the team needs to know in advance so they can hire, schedule, or triage accordingly. Getting this wrong means customers wait too long — and churn.

Specific Weekly forecast broken down by plan tier	Measurable Predicted ticket count, accuracy by MAE / RMSE	Achievable Historical ticket time series available for modeling	Relevant Operational readiness — directly affects launch quality	Time-bound 4-week forecast starting from launch date
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Data Science / ML

📄 support_tickets.csv

FEATURE ADOPTION & PRODUCT

Q6

Which product features show the highest and lowest adoption rates across beta phases, and does low feature adoption predict eventual churn?

Not all customers use every feature. Some features are used by almost everyone; others are barely touched. This question maps out which features are popular and which are

ignored — then asks: do customers who never use certain key features end up cancelling more often?

Why it matters

If a feature is both underused and linked to churn, it might need better onboarding or promotion. It could also signal the feature is not valuable and should be dropped before launch.

Specific

Feature-level adoption rates, by beta phase

Measurable

Adoption rate %, churn rate for non-adopters

Achievable

Feature usage log data available in the dataset

Relevant

Shapes product roadmap and onboarding before launch

Time-bound

Scoped to the beta period only

Data Analytics

 feature_usage data, churn_events.csv